future—that massive changes on the surface of Venus are about to break out again.

EVEN MORE UNEXPECTED than the great Martian volcanos or the magma-flooded surface of Venus is what awaited us when the *Voyager 1* spacecraft encountered Io, the innermost of the four large Galilean moons of Jupiter, in March 1979. There we found a strange, small, multihued world positively awash in volcanos. As we watched in astonishment, eight active plumes poured gas and fine particles up into the sky. The largest, now called Pelé—after the Hawaiian volcano goddess—projected a fountain of material 250 kilometers into space, higher above the surface of Io than some astronauts have ventured above the Earth. By the time *Voyager 2* arrived at Io, four months later, Pelé had turned itself off, although six of the other plumes were still active, at least one new plume had been discovered, and another caldera, named Surt, had changed its color dramatically.

The colors of Io, even though exaggerated in NASA's color-enhanced images, are like none elsewhere in the Solar System. The currently favored explanation is that the Ionian volcanos are driven not by upwelling molten rock, as on the Earth, the Moon, Venus, and Mars, but by upwelling sulfur dioxide and molten sulfur. The surface is covered with volcanic mountains, volcanic calderas, vents, and lakes of molten sulfur. Various forms and compounds of sulfur have been detected on the surface of Io and in nearby space—the volcanos blow some of the sulfur off Io altogether.* These findings have suggested to some an underground sea of liquid sulfur that issues to the surface at points of weakness, generates a shallow volcanic mound, trickles downhill, and freezes, its final color determined by its temperature on eruption.

On the Moon or Mars, you can find many places that have changed little in a billion years. On Io, in a century, much of the surface should be reflooded, filled in or washed away by new volcanic flows. Maps of Io will then quickly become obsolete, and cartography of Io will have become a growth industry.